



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

Experiment 5

PREPARATION OF AN ENZYME EXTRACT (POLYPHENOL OXIDASE -PPO)

Pre-Lab Questions:

- 1) Define the Enzyme.
- 2) What are the factors affect Polyphenol Oxidases?
- 3) Write two inhibitors of this enzyme.
- 4) What is the purpose of adding ascorbic acid (lemon) when washing some vegetables and fruits?

Precautions

- **Waste disposal:** Please dispose the liquid wastes in a large beaker
- The lab technician needs to adjust the water bath at required temperatures.
- The PPO enzyme will be ready for you after extracting it from potatoes.
- Catechol Phenol and hydroquinone are poisons; Avoid contact with solutions.

Objectives:

1. To demonstrate the chemical nature of the enzyme.
2. To demonstrate activity of the enzyme polyphenol oxidase in crude extract prepared from potato.
3. To investigate the substrate specificity of the enzyme
4. To investigate the effects of various factors on the activity of the enzyme.

Introduction

Enzymes are naturally occurring catalysts that are also known as biological catalyst, most of known enzymes are proteins. They are high molecular weight compounds made up principally of chains of amino acids linked together by peptide bonds. Therefore it is expected that every factor effect on protein will affect the enzyme relatively. These factors could be: concentration and type of certain substances, Temperature, pH value and the effect of inhibitors.

Polyphenol oxidase (tyrosinase) is a bifunctional, copper-containing oxidase enzyme. It is an oxygen transferring enzyme. Besides using O_2 to catalyze the dehydrogenation of catechols to orthoquinones and the orthohydroxylation of phenols to catechols (**figure 1**). This sort of reaction commonly occurs in nature and accounts for the browning of peeled potato, bananas, and bruished fruits.

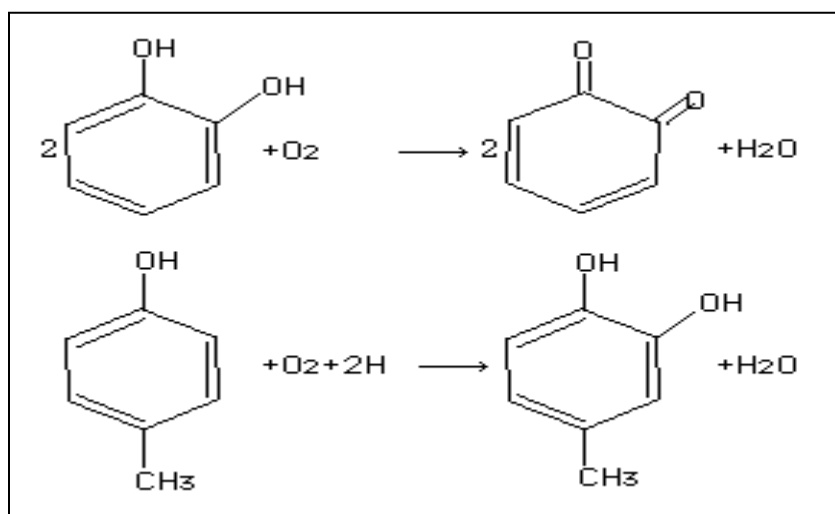


Figure 1. Polyphenol oxidase reaction

If Oxygen is removed from the immediate environment of the enzyme ,catalysis will be inhibited. Ascorbic acid (Vitamin C) can non-enzymatically scavenge oxygen, thereby lowering the rate of catalysis.

In this experiment, the PPO Enzyme will be extracted from potato in NaF solution. Then, you will study the factors that affecting the activity of this enzyme.

Experimental procedure

Materials needed

1. Phenol (0.1 M)
2. Catechol (0.01 M)
3. Hydroquinone (0.01M)
4. Hydrochloric acid (0.4%)
5. Lactic acid (0.1 %)
6. Sodium carbonate (0.5%)
7. Sodium Azide (0.1 M)
8. Distilled Water
9. Sodium fluoride (0.01 M)

Part A: Extraction of polyphenol oxidase.

Procedure:

1. Peel a reasonable sized potato.
2. Cut into pieces, place in a homogenizer.
3. Add 50 mL of 0.01 M NaF to prevent the oxidation of PPA with the oxygen of the air
4. Filter the extract is in a layer of cheese cloth and kept in a refrigerator till use.

Part B: Oxidation catalysis effect of polyphenol oxidase.

Procedure:

A) Substrate Specificity:

- 1- Label three test tubes A, B, and C.
- 2- Add 15 drops of enzyme extract to each test tube.
- 3- Prepare tubes as follows:
Tube A: 15 drops of 0.1 M Phenol.
Tube B: 15 drops of 0.01 M Catechol.
Tube C: 15 drops of 0.01 Hydroquinone.
- 4- Shake gently and place them in a water at 37 °C.
- 5- Examine the test tubes after 5 and 10 minute intervals and record the color change and its intensity using the color key for enzyme activity.

B) Substrate Concentration:

- 1- Label three test tubes A, B, and C.
- 2- Add 15 drops of enzyme extract to each test tube.

- 3- Prepare tubes as follows:
Tube A: 2 mL of 0.01 M Catechol and 4 mL distilled water.
Tube B: 4 mL of 0.01 M Catechol and 2 mL distilled water.
Tube C: 6 mL of 0.01 M Catechol.
- 4- Shake gently and place them in a water at 37 °C.
- 5- Examine the test tubes at 2-3 minute intervals for 20 minutes and record the color change and its intensity using the color key for enzyme activity.

C) Temperature effect on the enzyme action:

- 1- Label three test tubes A, B, and C.
- 2- Add 15 drops of enzyme extract to each test tube.
- 3- Prepare tubes as follows:
Tube A: 15 drops of 0.01 M Catechol.
Tube B: 15 drops of 0.01 M Catechol.
Tube C: 15 drops of 0.01 M catechol.
- 4- Place the first test tube at 0 °C, the second at 37 °C, and the third at 70 °C for 10 minutes each.
- 5- To each test tube, quickly add 15 drops of enzyme extract. Shake and return each to its proper temperature.
- 6- Examine each test tube after 15 minute for pertinent color change without removing the tubes from their temperature condition.
- 7- Record your results.

D) pH effect on the enzyme action:

- 1- Label four test tubes A, B, C, and D.
- 2- Prepare each test tube as follows.
Tube A: 2 mL of 0.4% Hydrochloric acid (pH=1).
Tube B: 2 mL of 0.1 % Lactic acid (pH=5).
Tube C: 2 mL of Distilled water.
Tube D: 2 mL of 0.5% Sodium carbonate.
- 3- Add 15 drops of 0.1 M Catechol to each tube.
- 4- Add 15 drops of enzyme extract to each tube.
- 5- Shake gently and place them in a water bath at 37 °C.
- 6- Wait for 15 minutes and examine each tube for pertinent color change.
- 7- Record your results.

E) Effect of inhibitors:

The activity of the enzyme (PPO) can be inhibited by the action of one of the following inhibitors:

Sodium Azide, Thiourea, Cinnamic Acid, L- Phenyl Alanine, and L- Tryptophan.

- 1- Add 15 drops of 0.1 M Sodium Azide or Thiourea to a tube containing 15 drops of the enzyme extract.

- 2- In another tube add 15 drops of enzyme extract.
- 3- Add 15 drops of 0.01 M Catechol to each tube.
- 4- Shake gently and keep in a water bath at 37 °C for 10 minutes.
- 5- Compare the color in both tubes and report your results.

Data analysis:

1. Write the report of this experiment in your own words.
2. Draw the relationship between color intensity and substrate status.

Make use of the following table, which shows the visual activity of polyphenol oxidases

Key	Degree in color intensity
-	No color Change
+	Light (faint) color
++	Definite color
+++	Dark (deep) color

References:

1. fac.ksu.edu.sa/sites/.../1-some_factors_affecting_polyphenol_oxidase_activity.pdf.
2. "Polyphenol Oxidase". *Worthington Enzyme Manual*. Retrieved 13 September 2011.
3. http://shodhganga.inflibnet.ac.in/bitstream/10603/100595/7/07_chapter%201.pdf



**Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 5**

Preparation of an enzyme extract (Polyphenol Oxidases (PPO))

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Purpose:

Data and Results:

A- Substrate Specificity

Substrate	Catechol	Phenol	Hydroquinone
Observation			

B- Effect of concentration

Concentration	6 mL Catechol	4 mL Catechol	2 mL Catechol
Observation			

C- Effect of Temperature

Temperature	0 °C	37 °C	70 °C
Observation			

D- Effect of pH

pH	1	5	7	9
Observation				

E-Effect of Inhibitors

	With Inhibitor	Without Inhibitor
Observation		

Conclusion:

Errors and problems you encountered and their possible remedies:

